北京大學

第五讲
高斯判别分析

机器学习概论

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内容

- 基本概念
- 参数学习
  - 最大似然估计
  - 最大后验估计
- 预测
- 最近类中心点分类器
- Fisher线性判别分析

基本概念

- 生成模型

$$p(y=c|\boldsymbol{x};\boldsymbol{\theta})=\frac{p(y=c;\boldsymbol{\theta})p(\boldsymbol{x}|y=c;\boldsymbol{\theta})}{p(\boldsymbol{x};\boldsymbol{\theta})}=\frac{p(y=c;\boldsymbol{\theta})p(\boldsymbol{x}|y=c;\boldsymbol{\theta})}{\sum_{c'=1}^{C}p(y=c';\boldsymbol{\theta})p(\boldsymbol{x}|y=c';\boldsymbol{\theta})}$$

  - $p(y=c;\boldsymbol{\theta})$： 类标签先验分布
  - $p(\boldsymbol{x}|y=c;\boldsymbol{\theta})$：类$c$的类条件分布；通过采样$p(\boldsymbol{x}|y=c;\boldsymbol{\theta})$，能够生成类别$c$ 的数据$\boldsymbol{x}$
  - $\boldsymbol{\theta}$：模型参数

- 高斯判别模型
  - $\boldsymbol{x}\in\mathbb{R}^D, p(\boldsymbol{x}|y=c;\boldsymbol{\theta})$为多元高斯分布

- 指定多元高斯分布为类条件分布

$$p(\boldsymbol{x}|y=c;\boldsymbol{\theta})=\mathcal{N}(\boldsymbol{\mu}_c,\Sigma_c)=\frac{1}{(2\pi)^{D/2}|\Sigma_c|^{1/2}}\exp\left[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma_c{}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)\right]$$

- 类后验分布
  - $\alpha_c=p(y=c;\boldsymbol{\theta})$

注意：高斯判别分析是生成模型

$$p(y=c|\boldsymbol{x};\boldsymbol{\theta})\propto p(y=c;\boldsymbol{\theta})p(\boldsymbol{x}|y=c;\boldsymbol{\theta})=\alpha_c\mathcal{N}(\boldsymbol{\mu}_c,\Sigma_c)$$

$$\log p(y=c|\boldsymbol{x};\boldsymbol{\theta})=\log\alpha_c-\frac{1}{2}\log|\Sigma_c|-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma_c{}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)-\frac{D}{2}\log 2\pi$$

- 给定训练数据集$\mathcal{D} = \{(\boldsymbol{x}_n, y_n)\}_{n=1}^{N}$，
  - $\boldsymbol{x}_n \in \mathbb{R}^D,\ y_n \in \{1, 2, \cdots, C\}$
- 模型参数$\boldsymbol{\theta} = \{\alpha_c\}_{c=1}^{C} \cup \{\boldsymbol{\mu}_c\}_{c=1}^{C} \cup \{\Sigma_c\}_{c=1}^{C}$
- 目标：求解$\boldsymbol{\theta}^*$，使得

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, p(\mathcal{D}; \boldsymbol{\theta})$$

  - 即，模型参数配置为$\boldsymbol{\theta}^*$时，训练样本集$\mathcal{D}$出现（生成）的概率最大
- 方法
  - 最大似然估计
  - 最大后验估计

最大似然估计—似然计算

$$
\begin{aligned}
p(\mathcal{D};\boldsymbol{\theta}) &= p(\{(\boldsymbol{x}_n, y_n)\}_{n=1}^{N};\boldsymbol{\theta}) \\
&= \prod_{n=1}^{N} p(\boldsymbol{x}_n, y_n;\boldsymbol{\theta}) \\
&= \prod_{n=1}^{N} p(y_n;\boldsymbol{\theta}) p(\boldsymbol{x}_n|y_n;\boldsymbol{\theta}) \\
&= \prod_{n=1}^{N} p(y_n;\boldsymbol{\theta}) \prod_{n=1}^{N} p(\boldsymbol{x}_n|y_n;\boldsymbol{\theta}) \\
&= \left(\prod_{n=1}^{N}\prod_{c=1}^{C} p(c;\boldsymbol{\theta})^{\mathbb{I}(y_n=c)}\right)\left(\prod_{n=1}^{N}\prod_{c=1}^{C}\big(p(\boldsymbol{x}_n|c;\boldsymbol{\theta})\big)^{\mathbb{I}(y_n=c)}\right)
\end{aligned}
$$

最大似然估计—似然计算

$$\boldsymbol{\theta} = \{\alpha_c\}_{c=1}^{C} \cup \{\boldsymbol{\mu}_c\}_{c=1}^{C} \cup \{\Sigma_c\}_{c=1}^{C}$$

$$p(\mathcal{D};\boldsymbol{\theta}) = \left(\prod_{n=1}^{N}\prod_{c=1}^{C} p(c;\boldsymbol{\theta})^{\mathbb{I}(y_n=c)}\right)\left(\prod_{n=1}^{N}\prod_{c=1}^{C}\left(p(\boldsymbol{x}_n|c;\boldsymbol{\theta})\right)^{\mathbb{I}(y_n=c)}\right)$$

$$= \left(\prod_{n=1}^{N}\prod_{c=1}^{C} {\alpha_c}^{\mathbb{I}(y_n=c)}\right)\left(\prod_{n=1}^{N}\prod_{c=1}^{C}\left(\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)\right)^{\mathbb{I}(y_n=c)}\right)$$

$$\log p(\mathcal{D};\boldsymbol{\theta}) = \sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n=c)\log\alpha_c + \sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n=c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

最大似然估计—负对数似然

$$\log p(\mathcal{D};\boldsymbol{\theta}) = \sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\log\alpha_c + \sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

$$= \sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n = c)\log\alpha_c + \sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n = c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

$$\mathrm{NLL}(\boldsymbol{\theta}) = -\log p(\mathcal{D};\boldsymbol{\theta})$$

$$= -\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n = c)\log\alpha_c - \sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n = c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

ITML 最大似然估计—求解

最大似然估计：$\max_{\boldsymbol{\theta}} p(\mathcal{D};\boldsymbol{\theta})$

$$\max_{\boldsymbol{\theta}} p(\mathcal{D};\boldsymbol{\theta}) \Leftrightarrow \max_{\boldsymbol{\theta}} \log p(\mathcal{D};\boldsymbol{\theta}) \Leftrightarrow \min_{\boldsymbol{\theta}} \mathrm{NLL}(\boldsymbol{\theta})$$

$$\mathrm{NLL}(\boldsymbol{\theta}) = -\sum_{c=1}^{C}\sum_{n=1}^{N} \mathbb{I}(y_n = c)\log \alpha_c - \sum_{c=1}^{C}\sum_{n=1}^{N} \mathbb{I}(y_n = c)\log \mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

约束条件： $\sum_{c=1}^{C} \alpha_c = 1$

拉格朗日乘子法：

$$\mathcal{L}(\boldsymbol{\theta},\lambda) = \mathrm{NLL}(\boldsymbol{\theta}) + \lambda\left(\sum_{c=1}^{C}\alpha_c - 1\right)$$

最大似然估计—求解

$$\boldsymbol{\theta}^* = \min_{\boldsymbol{\theta}} \mathrm{NLL}(\boldsymbol{\theta})$$

满足条件：

$$\left.\frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \lambda}\right|_{\lambda=\lambda^*} = 0$$

$$\left.\frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*} = \mathbf{0}$$

$$\Downarrow$$

计算

$$\begin{cases} \dfrac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \lambda} \\[2ex] \dfrac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\theta}} \end{cases}$$

最大似然估计—梯度计算

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \lambda} = \sum_{c=1}^{C} \alpha_c - 1$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\theta}} \Longrightarrow \begin{cases} \dfrac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \alpha_i}, 1 \le i \le C \\ \dfrac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\mu}_i}, 1 \le i \le C \\ \dfrac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \Sigma_i}, 1 \le i \le C \end{cases}$$

最大似然估计—梯度计算

$$\mathcal{L}(\boldsymbol{\theta},\lambda)=\mathrm{NLL}(\boldsymbol{\theta})+\lambda\left(\sum_{c=1}^{C}\alpha_c-1\right)\qquad \mathrm{NLL}(\boldsymbol{\theta})=-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\alpha_c-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

$$\frac{\partial\mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial\alpha_i}=\frac{\partial}{\partial\alpha_i}\left(-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\alpha_c\right)+\frac{\partial}{\partial\alpha_i}\left(\lambda\left(\sum_{c=1}^{C}\alpha_c-1\right)\right)$$

$$=\frac{\partial}{\partial\alpha_i}\left(-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\alpha_c\right)+\lambda$$

令$N_i=\sum_{n=1}^{N}\mathbb{I}(y_n=i)$

$$=\frac{\partial}{\partial\alpha_i}\left(-\sum_{n=1}^{N}\mathbb{I}(y_n=i)\log\alpha_i\right)+\lambda=-\sum_{n=1}^{N}\frac{\mathbb{I}(y_n=i)}{\alpha_i}+\lambda=\lambda-\frac{N_i}{\alpha_i}$$

最大似然估计—梯度计算

$$\mathcal{L}(\boldsymbol{\theta},\lambda) = \mathrm{NLL}(\boldsymbol{\theta}) + \lambda\left(\sum_{c=1}^{C}\alpha_c - 1\right) \qquad \mathrm{NLL}(\boldsymbol{\theta}) = -\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n = c)\log\alpha_c - \sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n = c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

$$\frac{\partial\mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial\boldsymbol{\mu}_i} = \frac{\partial}{\partial\boldsymbol{\mu}_i}\left(-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n = c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)\right)$$

$$= -\frac{\partial}{\partial\boldsymbol{\mu}_i}\left(\sum_{n=1}^{N}\mathbb{I}(y_n = i)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_i,\Sigma_i)\right)$$

$$= -\sum_{n=1}^{N}\mathbb{I}(y_n = i)\frac{\partial}{\partial\boldsymbol{\mu}_i}\left(\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_i,\Sigma_i)\right)$$

最大似然估计—梯度计算

$$\log \mathcal{N}(\boldsymbol{x}_n; \boldsymbol{\mu}_i, \Sigma_i) = -\frac{D}{2}\log 2\pi - \frac{1}{2}\log|\Sigma_i| - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\Sigma_i^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\mu}_i} = -\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \boldsymbol{\mu}_i}(\log \mathcal{N}(\boldsymbol{x}_n; \boldsymbol{\mu}_i, \Sigma_i))$$

$$= -\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \boldsymbol{\mu}_i}\left(-\frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\Sigma_i^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)\right)$$

两类计算方法

$$= \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \boldsymbol{\mu}_i}\left((\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\Sigma_i^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)\right)$$

最大似然估计—梯度计算

计算方法一

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \boldsymbol{\mu}_i} = \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \boldsymbol{\mu}_i}\left((\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\Sigma_i{}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)\right)$$

$$= \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \boldsymbol{\mu}_i}\left(\boldsymbol{x}_n{}^{\mathrm{T}}\Sigma_i{}^{-1}\boldsymbol{x}_n - 2\boldsymbol{x}_n{}^{\mathrm{T}}\Sigma_i{}^{-1}\boldsymbol{\mu}_i + \boldsymbol{\mu}_i{}^{\mathrm{T}}\Sigma_i{}^{-1}\boldsymbol{\mu}_i\right)$$

$$\frac{\partial \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x}}{\partial \boldsymbol{x}} = \boldsymbol{a} \qquad \frac{\partial \boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}}{\partial \boldsymbol{x}} = (\mathbf{A} + \mathbf{A}^{\mathrm{T}})\boldsymbol{x}$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \boldsymbol{\mu}_i} = \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\left(-2(\Sigma_i{}^{-1})^{\mathrm{T}}\boldsymbol{x}_n + \left(\Sigma_i{}^{-1} + (\Sigma_i{}^{-1})^{\mathrm{T}}\right)\boldsymbol{\mu}_i\right)$$

$$\Sigma_i = \Sigma_i{}^{\mathrm{T}} \Rightarrow \Sigma_i{}^{-1} = (\Sigma_i{}^{-1})^{\mathrm{T}}$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \boldsymbol{\mu}_i} = \sum_{n=1}^{N} \mathbb{I}(y_n = i)\Sigma_i{}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{x}_n)$$

最大似然估计—梯度计算

计算方法二

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \boldsymbol{\mu}_i} = \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \boldsymbol{\mu}_i}\left((\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\Sigma_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)\right)$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \boldsymbol{\mu}_i} = \frac{\partial (\boldsymbol{x}_n - \boldsymbol{\mu}_i)}{\partial \boldsymbol{\mu}_i}\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial (\boldsymbol{x}_n - \boldsymbol{\mu}_i)}$$

$$\frac{\partial \boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}}{\partial \boldsymbol{x}} = \left(\mathbf{A} + \mathbf{A}^{\mathrm{T}}\right)\boldsymbol{x}$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \boldsymbol{\mu}_i} = \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)(-\mathbf{I})\left({\Sigma_i}^{-1} + ({\Sigma_i}^{-1})^{\mathrm{T}}\right)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)$$

$$\Sigma_i = {\Sigma_i}^{\mathrm{T}} \Rightarrow {\Sigma_i}^{-1} = ({\Sigma_i}^{-1})^{\mathrm{T}}$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \boldsymbol{\mu}_i} = \sum_{n=1}^{N} \mathbb{I}(y_n = i){\Sigma_i}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{x}_n)$$

最大似然估计—梯度计算

$$\mathcal{L}(\boldsymbol{\theta},\lambda)=\mathrm{NLL}(\boldsymbol{\theta})+\lambda\left(\sum_{c=1}^{C}\alpha_c-1\right)\qquad \mathrm{NLL}(\boldsymbol{\theta})=-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\alpha_c-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)$$

$$\frac{\partial\mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial\Sigma_i}=\frac{\partial}{\partial\Sigma_i}\left(-\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)\right)$$

$$=\frac{\partial}{\partial\Sigma_i}\left(-\sum_{n=1}^{N}\mathbb{I}(y_n=i)\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_i,\Sigma_i)\right)$$

$$=-\sum_{n=1}^{N}\mathbb{I}(y_n=i)\frac{\partial}{\partial\Sigma_i}\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_i,\Sigma_i)$$

ITML

最大似然估计—梯度计算

$$\log \mathcal{N}(\boldsymbol{x}_n; \boldsymbol{\mu}_i, \Sigma_i) = -\frac{D}{2}\log 2\pi - \frac{1}{2}\log|\Sigma_i| - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\Sigma_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \Sigma_i} = -\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \Sigma_i}\log \mathcal{N}(\boldsymbol{x}_n; \boldsymbol{\mu}_i, \Sigma_i)$$

$$= \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\frac{\partial}{\partial \Sigma_i}\left(\log|\Sigma_i| + (\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\Sigma_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)\right)$$

$$= \frac{1}{2}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\left(\frac{\partial \log|\Sigma_i|}{\partial \Sigma_i} + \frac{\partial (\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\Sigma_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)}{\partial \Sigma_i}\right)$$

?

最大似然估计—梯度计算

$$\frac{\partial \log|\mathbf{X}|}{\partial \mathbf{X}} = \mathbf{X}^{-\mathrm{T}}$$

$$\Downarrow$$

$$\frac{\partial \log|\Sigma_i|}{\partial \Sigma_i} = {\Sigma_i}^{-\mathrm{T}}$$

$$\mathbf{X}^{-\mathrm{T}} = (\mathbf{X}^{-1})^{\mathrm{T}} = (\mathbf{X}^{\mathrm{T}})^{-1}$$

$$\Downarrow$$

$$\frac{\partial \log|\Sigma_i|}{\partial \Sigma_i} = ({\Sigma_i}^{\mathrm{T}})^{-1} \xrightarrow{\Sigma_i = {\Sigma_i}^{\mathrm{T}}} \frac{\partial \log|\Sigma_i|}{\partial \Sigma_i} = {\Sigma_i}^{-1}$$

最大似然估计—梯度计算

$$\mathrm{tr}(\mathbf{ABC}) = \mathrm{tr}(\mathbf{BCA}) = \mathrm{tr}(\mathbf{CAB})$$

$$\begin{aligned}
&(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i) \\
&= \mathrm{tr}\left((\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)\right) \\
&= \mathrm{tr}\left((\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-1}\right) \\
&= \mathrm{tr}\left({\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\right)
\end{aligned}$$

$$\frac{\partial(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)}{\partial{\textstyle\sum_i}} = \frac{\partial\mathrm{tr}\left({\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\right)}{\partial{\textstyle\sum_i}}$$

$$\frac{\partial\mathrm{tr}(\mathbf{X}^{-1}\mathbf{A})}{\partial\mathbf{X}} = -\mathbf{X}^{-\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{X}^{-\mathrm{T}}$$

$$= -{\textstyle\sum_i}^{-\mathrm{T}}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-\mathrm{T}}$$

$$\mathbf{X}^{-\mathrm{T}} = (\mathbf{X}^{-1})^{\mathrm{T}} = (\mathbf{X}^{\mathrm{T}})^{-1} \qquad {\textstyle\sum_i} = {\textstyle\sum_i}^{\mathrm{T}}$$

$$= -{\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-1}$$

$$\frac{\partial\log|{\textstyle\sum_i}|}{\partial{\textstyle\sum_i}} = {\textstyle\sum_i}^{-1}$$

$$\begin{aligned}
&\frac{\partial\mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial{\textstyle\sum_i}} \\
&= \frac{1}{2}\sum_{n=1}^{N}\mathbb{I}(y_n = i)\frac{\partial}{\partial{\textstyle\sum_i}}\left(\frac{\partial\log|{\textstyle\sum_i}|}{\partial{\textstyle\sum_i}} + \frac{\partial(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)}{\partial{\textstyle\sum_i}}\right) \\
&= \frac{1}{2}\sum_{n=1}^{N}\mathbb{I}(y_n = i)\left({\textstyle\sum_i}^{-1} - {\textstyle\sum_i}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}{\textstyle\sum_i}^{-1}\right)
\end{aligned}$$

最大似然估计—参数求解

$$\left.\begin{aligned} 0 &= \frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \lambda} = \sum_{c=1}^{C} \alpha_c - 1 \\ 0 &= \frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \alpha_i} = \lambda - \frac{N_i}{\alpha_i} \end{aligned}\right\} \Rightarrow \alpha_i = \frac{N_i}{\sum_{c=1}^{C} N_c} = \frac{N_i}{N}$$

$$0 = \frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\mu}_i} = \sum_{n=1}^{N} \mathbb{I}(y_n = i) {\Sigma_i}^{-1} (\boldsymbol{\mu}_i - \boldsymbol{x}_n)$$

$$0 = \frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \Sigma_i} = \frac{1}{2} \sum_{n=1}^{N} \mathbb{I}(y_n = i) \left({\Sigma_i}^{-1} - {\Sigma_i}^{-1} (\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}} {\Sigma_i}^{-1} \right)$$

最大似然估计—参数求解

$$0 = \frac{\partial \mathcal{L}(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\mu}_i} = -\sum_{n=1}^{N} \mathbb{I}(y_n = i)\Sigma_i^{-1}(\boldsymbol{\mu}_i - \boldsymbol{x}_n)$$

$$\Downarrow$$

$$\Sigma_i^{-1}\left(\sum_{n=1}^{N} \mathbb{I}(y_n = i)(\boldsymbol{\mu}_i - \boldsymbol{x}_n)\right) = 0$$

$$\Downarrow$$

$$\sum_{n=1}^{N} \mathbb{I}(y_n = i)(\boldsymbol{\mu}_i - \boldsymbol{x}_n) = 0$$

$$\Rightarrow$$

$$\sum_{n=1}^{N} \mathbb{I}(y_n = i)\boldsymbol{\mu}_i = \sum_{n=1}^{N} \mathbb{I}(y_n = i)\boldsymbol{x}_n$$

$$\Uparrow$$

$$\boldsymbol{\mu}_i = \frac{1}{N_i}\sum_{n=1}^{N} \mathbb{I}(y_n = i)\boldsymbol{x}_n = \frac{1}{N_i}\sum_{n:y_n=i} \boldsymbol{x}_n$$

最大似然估计—参数求解

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta},\lambda)}{\partial \Sigma_i}=\frac{1}{2}\sum_{n=1}^{N}\mathbb{I}(y_n=i)\left(\Sigma_i^{-1}-\Sigma_i^{-1}(\boldsymbol{x}_n-\boldsymbol{\mu}_i)(\boldsymbol{x}_n-\boldsymbol{\mu}_i)^{\mathrm{T}}\Sigma_i^{-1}\right)=0$$

$$\sum_{n=1}^{N}\mathbb{I}(y_n=i)\,\Sigma_i^{-1}=\sum_{n=1}^{N}\mathbb{I}(y_n=i)\,\Sigma_i^{-1}(\boldsymbol{x}_n-\boldsymbol{\mu}_i)(\boldsymbol{x}_n-\boldsymbol{\mu}_i)^{\mathrm{T}}\Sigma_i^{-1}$$

$$\Sigma_i\left(\sum_{n=1}^{N}\mathbb{I}(y_n=i)\,\Sigma_i^{-1}\right)\Sigma_i=\Sigma_i\left(\sum_{n=1}^{N}\mathbb{I}(y_n=i)\,\Sigma_i^{-1}(\boldsymbol{x}_n-\boldsymbol{\mu}_i)(\boldsymbol{x}_n-\boldsymbol{\mu}_i)^{\mathrm{T}}\Sigma_i^{-1}\right)\Sigma_i$$

$$\sum_{n=1}^{N}\mathbb{I}(y_n=i)\,\Sigma_i=\sum_{n=1}^{N}\mathbb{I}(y_n=i)\,(\boldsymbol{x}_n-\boldsymbol{\mu}_i)(\boldsymbol{x}_n-\boldsymbol{\mu}_i)^{\mathrm{T}} \Rightarrow \Sigma_i=\frac{1}{N_i}\sum_{n:y_n=i}(\boldsymbol{x}_n-\boldsymbol{\mu}_i)(\boldsymbol{x}_n-\boldsymbol{\mu}_i)^{\mathrm{T}}$$

最大似然估计—讨论

- 协方差矩阵估计

$$\Sigma_i = \frac{1}{N_i} \sum_{n:y_n=i} (\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}$$

  - 当$N_i < D$，即标注为类别i的样本数据量小于数据维度时， Σ_i 会退化成半正定矩阵，此时估计就不是良态（well-conditioned），与多元高斯分布中协方差矩阵（实数域）是正定矩阵定义相悖

    ✓注意：正定矩阵是可逆矩阵，半正定矩阵不一定是可逆矩阵

  - 原因探究：当$N_i < D$，意味着大参数模型去拟合小数据，容易过拟合（Overfitting）

ITML 最大似然估计—讨论

- $N_i < D$时，估计的协方差矩阵为半正定矩阵

$$\Sigma_i = \frac{1}{N_i}\sum_{n:y_n=i}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}$$

- 证明：存在D维非零向量$\boldsymbol{a}(\neq \boldsymbol{0})$，满足$\boldsymbol{a}^{\mathrm{T}}\Sigma_i\boldsymbol{a} = 0$

$$\boldsymbol{a}^{\mathrm{T}}\Sigma_i\boldsymbol{a} = \frac{1}{N_i}\sum_{n:y_n=i}\boldsymbol{a}^{\mathrm{T}}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\boldsymbol{a} = \frac{1}{N_i}\sum_{n:y_n=i}\left((\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\boldsymbol{a}\right)^{\mathrm{T}}\left((\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}\boldsymbol{a}\right)$$

- 为处理方便，令$K = N_i$，并记$\{\boldsymbol{x}_n - \boldsymbol{\mu}_i | y_n = i; 0 \leq n \leq N\} = \{\boldsymbol{z}_k\}(0 \leq k \leq K)$

$$\boldsymbol{a}^{\mathrm{T}}\Sigma_i\boldsymbol{a} = \frac{1}{K}\sum_{k=1}^{K}\left(\boldsymbol{z}_k{}^{\mathrm{T}}\boldsymbol{a}\right)^{\mathrm{T}}\left(\boldsymbol{z}_k{}^{\mathrm{T}}\boldsymbol{a}\right) = \frac{1}{K}\sum_{k=1}^{K}\left(\boldsymbol{z}_k{}^{\mathrm{T}}\boldsymbol{a}\right)^2$$

最大似然估计—讨论

$$\boldsymbol{a}^{\mathrm{T}}\Sigma_i\boldsymbol{a}=\mathbf{0}\Leftrightarrow\frac{1}{K}\sum_{k=1}^{K}\left(\boldsymbol{z}_k{}^{\mathrm{T}}\boldsymbol{a}\right)^2=\mathbf{0}\Leftrightarrow\begin{cases}\left(\boldsymbol{z}_1{}^{\mathrm{T}}\boldsymbol{a}\right)^2=0\\ \quad\vdots\\ \left(\boldsymbol{z}_K{}^{\mathrm{T}}\boldsymbol{a}\right)^2=0\end{cases}\Leftrightarrow\begin{cases}\boldsymbol{z}_1{}^{\mathrm{T}}\boldsymbol{a}=0\\ \quad\vdots\\ \boldsymbol{z}_K{}^{\mathrm{T}}\boldsymbol{a}=0\end{cases}$$

$$\begin{cases}\boldsymbol{z}_1{}^{\mathrm{T}}\boldsymbol{a}=0\\ \quad\vdots\\ \boldsymbol{z}_K{}^{\mathrm{T}}\boldsymbol{a}=0\end{cases}\xrightarrow{\mathbf{Z}=[\boldsymbol{z}_1,\cdots,\boldsymbol{z}_K]}\mathbf{Z}^{\mathrm{T}}\boldsymbol{a}=\mathbf{0}$$

$\mathbf{Z}^{\mathrm{T}}$是$K\times D$的矩阵，$\boldsymbol{a}$是D维向量。
当$K<D$，上述等式有无穷个非零解

最大似然估计—共享协方差矩阵

- 在训练样本数量不变的条件下，通过减少模型参数量来减轻过学习
- 对应各类别的高斯分布共享相同的协方差矩阵
  - $\Sigma_i = \Sigma$
- 通过最大似然估计求解可得

$$\Sigma = \frac{1}{N}\sum_{i=1}^{C}\sum_{n:y_n=i}(\boldsymbol{x}_n - \boldsymbol{\mu}_i)(\boldsymbol{x}_n - \boldsymbol{\mu}_i)^{\mathrm{T}}$$

  - 通常，$N > D$。因此，上述估计协方差矩阵的方法更为可靠
- 这类高斯判别分析也称为线性判别分析（LDA, Linear Discriminant Analysis）

- 为什么称为线性判别分析

$$p(y=c|\boldsymbol{x};\boldsymbol{\theta})=\frac{p(y=c;\boldsymbol{\theta})p(\boldsymbol{x}|y=c;\boldsymbol{\theta})}{p(\boldsymbol{x};\boldsymbol{\theta})}=\frac{p(y=c;\boldsymbol{\theta})p(\boldsymbol{x}|y=c;\boldsymbol{\theta})}{p(\boldsymbol{x})}$$

$$\log p(y=c|\boldsymbol{x};\boldsymbol{\theta})=\log p(y=c;\boldsymbol{\theta})+\log p(\boldsymbol{x}|y=c;\boldsymbol{\theta})-\log p(\boldsymbol{x})$$

$$p(\boldsymbol{x}|y=c;\boldsymbol{\theta})=\mathcal{N}(\boldsymbol{\mu}_c,\Sigma)=\frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}}\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)\right)$$

$$\log p(y=c|\boldsymbol{x};\boldsymbol{\theta})=\log\alpha_c-\frac{D}{2}\log 2\pi-\frac{1}{2}\log|\Sigma|-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)-\log p(\boldsymbol{x})$$

$$=\underbrace{{\boldsymbol{\mu}_c}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{x}}_{\boldsymbol{x}\text{的线性函数}}+\underbrace{\log\alpha_c-\frac{1}{2}{\boldsymbol{\mu}_c}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{\mu}_c}_{\text{对给定类别}c\text{是常量}}\underbrace{-\frac{D}{2}\log 2\pi-\frac{1}{2}\log|\Sigma|-\frac{1}{2}\boldsymbol{x}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{x}-\log p(\boldsymbol{x})}_{\text{对所有类别相同}}$$

最大似然估计—对角协方差矩阵

- 减少参数量还有其他方法
  - 建模类协方差矩阵为对角矩阵，即

$$\Sigma_i = \begin{bmatrix} \sigma_1^i & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_D^i \end{bmatrix}$$

✓优点：参数量由$O(CD^2)$减少至$O(CD)$，可减轻过学习，适合建模高维数据

✓缺点：无法表达（刻画）数据特征之间的关联

  - 建模所有的类协方差矩阵为相同的对角矩阵，即

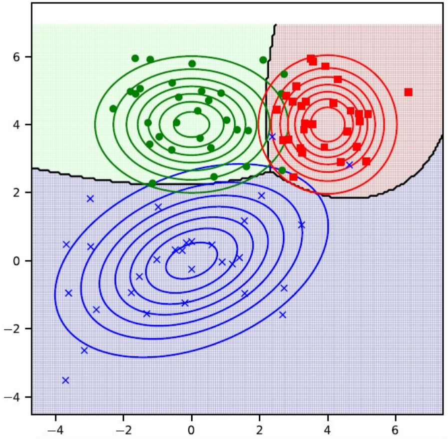
$$\Sigma_i = \begin{bmatrix} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_D \end{bmatrix}$$

✓进一步限制模型容量，参数量由$O(CD^2)$减少至$O(D)$

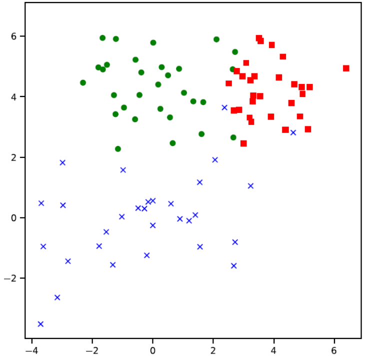
决策边界

- 高斯判别分析
  - 二次决策边界

$$\log p(y=c|\boldsymbol{x};\boldsymbol{\theta})$$
$$=\log\alpha_c-\frac{D}{2}\log 2\pi-\frac{1}{2}\log|\Sigma_c|-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}{\Sigma_c}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)$$

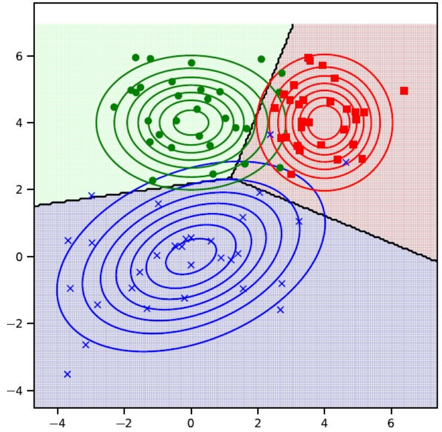


$$\log p(y=c_1|\boldsymbol{x};\boldsymbol{\theta})-\log p(y=c_2|\boldsymbol{x};\boldsymbol{\theta})=0$$



- 线性判别分析
  - 线性决策边界

$$\log p(y=c|\boldsymbol{x};\boldsymbol{\theta})$$
$$=\boldsymbol{\mu}_c{}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{x}+\log\alpha_c-\frac{1}{2}\boldsymbol{\mu}_c{}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{\mu}_c-\frac{D}{2}\log 2\pi-\frac{1}{2}\log|\Sigma|-\frac{1}{2}\boldsymbol{x}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{x}-\log p(\boldsymbol{x})$$



最大后验估计

- 最大似然估计
  - 问题：在处理高维数据的时候，估计出的协方差矩阵通常是奇异矩阵（确切的说是半正定矩阵）
  - 虽然可以通过协方差矩阵共享和对角化技术克服上述问题，但是这些技术包含相当强的假设条件
- 最大后验估计
  - 视协方差矩阵为随机变量，服从某个概率分布
  - 避免估计的协方差矩阵是奇异矩阵

$$\boldsymbol{\theta} = \underbrace{\{\alpha_c\}_{c=1}^{C} \cup \{\boldsymbol{\mu}_c\}_{c=1}^{C}}_{\text{未知变量(非随机变量)}} \cup \underset{\text{随机变量}}{\underset{\downarrow}{\{\Sigma_c\}_{c=1}^{C}}}$$

令$\boldsymbol{\alpha} = \{\alpha_c\}_{c=1}^{C}$,$\boldsymbol{\mu} = \{\boldsymbol{\mu}_c\}_{c=1}^{C}$。假定$\Sigma_c = \Sigma$，$1 \le c \le C$；且服从如下先验分布（逆威沙特分布，inverse Wishart distribution），其中$\mathbf{S}$为$D \times D$正定矩阵，ν是自由度

超参数，预先设定

$$\mathrm{IW}(\Sigma; \mathbf{S}, \nu) = \frac{1}{\underset{\text{归一化因子}}{\underset{\downarrow}{Z_{IW}}}} |\Sigma|^{-\frac{\nu+D+1}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}\left(\mathbf{S}^{-1}\Sigma^{-1}\right)\right)$$

最大后验估计—似然计算

$$p(\Sigma|\mathcal{D};\boldsymbol{\alpha},\boldsymbol{\mu})=\frac{p(\mathcal{D}|\Sigma;\boldsymbol{\alpha},\boldsymbol{\mu})p(\Sigma;\boldsymbol{\alpha},\boldsymbol{\mu})}{p(\mathcal{D};\boldsymbol{\alpha},\boldsymbol{\mu})}$$

$$=\frac{p(\mathcal{D}|\Sigma;\boldsymbol{\alpha},\boldsymbol{\mu})p(\Sigma;\boldsymbol{\alpha},\boldsymbol{\mu})}{p(\mathcal{D};\boldsymbol{\alpha},\boldsymbol{\mu})}$$

$$=\frac{p(\mathcal{D}|\Sigma;\boldsymbol{\alpha},\boldsymbol{\mu})p(\Sigma)}{p(\mathcal{D};\boldsymbol{\alpha},\boldsymbol{\mu})}$$

$$=\frac{1}{p(\mathcal{D};\boldsymbol{\alpha},\boldsymbol{\mu})}\left(\prod_{n=1}^{N}p(\boldsymbol{x}_n,y_n;\boldsymbol{\alpha},\boldsymbol{\mu},\Sigma)\right)\mathrm{IW}(\Sigma;\mathbf{S},\nu)$$

$$=\frac{1}{p(\mathcal{D};\boldsymbol{\alpha},\boldsymbol{\mu})}\left(\prod_{n=1}^{N}\prod_{c=1}^{C}\alpha_c{}^{\mathbb{I}(y_n=c)}\right)\left(\prod_{n=1}^{N}\prod_{c=1}^{C}\left(\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)\right)^{\mathbb{I}(y_n=c)}\right)\mathrm{IW}(\Sigma;\mathbf{S},\nu)$$

最大后验估计—梯度计算（Σ）

$$\max_{\Sigma} p(\Sigma|\mathcal{D};\boldsymbol{\alpha},\boldsymbol{\mu})$$

$$\Updownarrow$$

$$\max_{\Sigma}\left(\prod_{n=1}^{N}\prod_{c=1}^{C}\left(\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma_c)\right)^{\mathbb{I}(y_n=c)}\right)\mathrm{IW}(\Sigma;\mathbf{S},\nu)$$

$$\Updownarrow$$

$$\max_{\Sigma} Map(\Sigma) \triangleq \sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n=c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma)+\log\mathrm{IW}(\Sigma;\mathbf{S},\nu)$$

$$\Downarrow$$

$$\nabla_{\Sigma} Map(\Sigma)=\frac{\partial Map(\Sigma)}{\partial\Sigma}=\frac{\partial}{\partial\Sigma}\left(\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n=c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma)\right)+\frac{\partial}{\partial\Sigma}\left(\log\mathrm{IW}(\Sigma;\mathbf{S},\nu)\right)=0$$

最大后验估计—梯度计算（∑）

$$\log \mathcal{N}(\boldsymbol{x}_n; \boldsymbol{\mu}_c, \Sigma) = -\frac{D}{2}\log 2\pi - \frac{1}{2}\log|\Sigma| - \frac{1}{2}(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_c)$$

$$\frac{\partial}{\partial \Sigma}\left(\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\log \mathcal{N}(\boldsymbol{x}_n; \boldsymbol{\mu}_c, \Sigma)\right)$$

$$= -\frac{1}{2}\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\left(\frac{\partial \log|\Sigma|}{\partial \Sigma} + \frac{\partial(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_c)}{\partial \Sigma}\right)$$

$$= -\frac{1}{2}\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\left(\Sigma^{-\mathrm{T}} - \Sigma^{-\mathrm{T}}(\boldsymbol{x}_n - \boldsymbol{\mu}_c)(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-\mathrm{T}}\right)$$

$$= -\frac{1}{2}\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\left(\Sigma^{-1} - \Sigma^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_c)(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}\right)$$

最大后验估计—梯度计算（∑）

$$\mathrm{IW}(\Sigma;\mathbf{S},\nu)=\frac{1}{Z_{IW}}|\Sigma|^{-\frac{\nu+D+1}{2}}\exp\left(-\frac{1}{2}\mathrm{tr}(\mathbf{S}^{-1}\Sigma^{-1})\right)$$

$$\frac{\partial}{\partial\Sigma}\left(\log\mathrm{IW}(\Sigma;\mathbf{S},\nu)\right)=-\frac{\partial}{\partial\Sigma}\left(\frac{\nu+D+1}{2}\log|\Sigma|+\frac{1}{2}\mathrm{tr}(\mathbf{S}^{-1}\Sigma^{-1})\right)$$

$$\mathrm{tr}(\mathbf{AB})=\mathrm{tr}\left(\mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\right)$$

$\mathbf{S},\Sigma$是对称矩阵

$$=-\frac{\nu+D+1}{2}\frac{\partial\log|\Sigma|}{\partial\Sigma}-\frac{1}{2}\frac{\partial\mathrm{tr}(\Sigma^{-1}\mathbf{S}^{-1})}{\partial\Sigma}$$

$\mathbf{S},\Sigma$是对称矩阵

$$\frac{\partial\mathrm{tr}(\mathbf{X}^{-1}\mathbf{A})}{\partial\mathbf{X}}=-\mathbf{X}^{-\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{X}^{-\mathrm{T}}$$

$$=-\frac{\nu+D+1}{2}\Sigma^{-1}+\frac{1}{2}\Sigma^{-1}\mathbf{S}^{-1}\Sigma^{-1}$$

最大后验估计—求解

$$\frac{\partial Map(\Sigma)}{\partial \Sigma} = \mathbf{0}$$

$$\Downarrow$$

$$-\frac{\nu + D + 1}{2}\Sigma^{-1} + \frac{1}{2}\Sigma^{-1}\mathbf{S}^{-1}\Sigma^{-1} - \frac{1}{2}\sum_{n=1}^{N}\sum_{i=1}^{C}\mathbb{I}(y_n = c)\left(\Sigma^{-1} - \Sigma^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_c)(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}\right) = \mathbf{0}$$

$$\Downarrow$$

$$\frac{\nu + D + 1}{2}\Sigma^{-1} - \frac{1}{2}\Sigma^{-1}\mathbf{S}^{-1}\Sigma^{-1} + \frac{1}{2}\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\left(\Sigma^{-1} - \Sigma^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu}_c)(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}\right) = \mathbf{0}$$

$$\Downarrow$$

$$\Sigma_{\mathrm{map}} = \Sigma = \frac{1}{N + \nu + D + 1}\left(\mathbf{S}^{-1} + \sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\,(\boldsymbol{x}_n - \boldsymbol{\mu}_c)(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\right)$$

最大后验估计—讨论

在共享类协方差矩阵（线性判别分析）的条件，可以推出最大似然估计Σ_{mle}如下

$$\Sigma_{\text{mle}} = \frac{1}{N}\left(\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\,(\boldsymbol{x}_n - \boldsymbol{\mu}_c)(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\right)$$

先验分布IW$(\Sigma;\mathbf{S},\nu)$的众数（mode）为$\frac{\mathbf{S}^{-1}}{\nu+D+1}$，简记为$\Sigma_0$，则

$$\Sigma_{\text{map}} = \frac{\nu+D+1}{N+\nu+D+1}\frac{\mathbf{S}^{-1}}{\nu+D+1} + \frac{N}{N+\nu+D+1}\frac{1}{N}\left(\sum_{n=1}^{N}\sum_{c=1}^{C}\mathbb{I}(y_n = c)\,(\boldsymbol{x}_n - \boldsymbol{\mu}_c)(\boldsymbol{x}_n - \boldsymbol{\mu}_c)^{\mathrm{T}}\right)$$

$$\lambda = \frac{\nu+D+1}{N+\nu+D+1}$$

$$\Sigma_{\text{map}} = \lambda\Sigma_0 + (1-\lambda)\Sigma_{\text{mle}}$$

最大后验估计—讨论

先验散度矩阵S可视情况灵活设置，通常采用如下形式

$$\mathbf{S} = (\nu + D + 1)\,\mathrm{diag}\,\Sigma_{\mathrm{mle}}$$

此时，最大后验估计的协方差矩阵中的元素为

$$\Sigma_{\mathrm{map}}(i,j) = \begin{cases} \Sigma_{\mathrm{mle}}(i,j) & i = j \\ (1-\lambda)\Sigma_{\mathrm{mle}}(i,j) & i \neq j \end{cases}$$

高斯判别分析—预测（分类）

- 给定训练数据集$\mathcal{D}=\{(\boldsymbol{x}_n, y_n)\}_{n=1}^{N}$，通过最大似然（或最大后验）估计出模型参数

$$\alpha_i=\frac{N_i}{\sum_{c=1}^{C}N_c}=\frac{N_i}{N} \qquad \boldsymbol{\mu}_i=\frac{1}{N_i}\sum_{n:y_n=i}\boldsymbol{x}_n \qquad \Sigma_i=\frac{1}{N_i}\sum_{n:y_n=i}(\boldsymbol{x}_n-\boldsymbol{\mu}_i)(\boldsymbol{x}_n-\boldsymbol{\mu}_i)^{\mathrm{T}}$$

- 对类别未知数据$\boldsymbol{x}$，根据贝叶斯决策理论，如下判定其类别$\hat{y}$

$$\hat{y}=\underset{c}{\operatorname{argmax}}\, p(y=c|\boldsymbol{x};\{\alpha_i\},\{\boldsymbol{\mu}_i\},\{\Sigma_i\}) = \underset{c}{\operatorname{argmax}}\left(\log\alpha_c-\frac{1}{2}\log|\Sigma_c|-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma_c{}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)\right)$$

$$=\underset{c}{\operatorname{argmax}}\frac{p(y=c,\boldsymbol{x};\{\alpha_i\},\{\boldsymbol{\mu}_i\},\{\Sigma_i\})}{p(\boldsymbol{x};\{\alpha_i\},\{\boldsymbol{\mu}_i\},\{\Sigma_i\})}$$

$$=\underset{c}{\operatorname{argmax}}\, p(y=c,\boldsymbol{x};\{\alpha_i\},\{\boldsymbol{\mu}_i\},\{\Sigma_i\}) \qquad =\underset{c}{\operatorname{argmax}}\log\left(\alpha_c\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu}_c,\Sigma_c)\right)$$

$$=\underset{c}{\operatorname{argmax}}\, p(y=c;\{\alpha_i\},\{\boldsymbol{\mu}_i\},\{\Sigma_i\})p(\boldsymbol{x}|y=c,\{\alpha_i\},\{\boldsymbol{\mu}_i\},\{\Sigma_i\})=\underset{c}{\operatorname{argmax}}\,\alpha_c\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu}_c,\Sigma_c)$$

最近类中心点分类器

- Nearest centroid classifier (或nearest class mean classifier, NCM)
- 在最大似然估计中，如果假定类的先验分布服从均匀分布，即$p(c)=\frac{1}{C}$，且共享类协方差矩阵Σ，则所求解的最优化问题变成

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \operatorname{NLL}(\boldsymbol{\theta})$$

$$\operatorname{NLL}(\boldsymbol{\theta}) = -\sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\frac{1}{C} - \sum_{c=1}^{C}\sum_{n=1}^{N}\mathbb{I}(y_n=c)\log\mathcal{N}(\boldsymbol{x}_n;\boldsymbol{\mu}_c,\Sigma) \qquad \boldsymbol{\theta} = \{\boldsymbol{\mu}_c\}_{c=1}^{C} \cup \{\Sigma\}$$

- 可求解出

$$\boldsymbol{\mu}_c = \frac{1}{N_c}\sum_{n:y_n=c}\boldsymbol{x}_n \qquad \Sigma = \frac{1}{N}\sum_{i=1}^{C}\sum_{n:y_n=i}(\boldsymbol{x}_n-\boldsymbol{\mu}_i)(\boldsymbol{x}_n-\boldsymbol{\mu}_i)^{\mathrm{T}}$$

- 对类别未知数据$\boldsymbol{x}$，如下判定其类别$\hat{y}$

Mahalanobis Distance

$\boldsymbol{x}$与类c中心点$\boldsymbol{\mu}_c$的马氏距离平方

$$\hat{y} = \underset{c}{\operatorname{argmax}}\left(-\frac{1}{2}\log|\Sigma| - \frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)\right) = \underset{c}{\operatorname{argmin}}(\boldsymbol{x}-\boldsymbol{\mu}_c)^{\mathrm{T}}\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_c)$$

最近类中心点分类器

- 最近类中心点分类器实际上是学习出度量距离的方法。因此，可以采用不同的距离度量方法，比如可以用欧氏距离，闵氏距离等不同距离度量来计算数据与类中心点的距离

$$\hat{y} = \underset{c}{\operatorname{argmin}} (\boldsymbol{x} - \boldsymbol{\mu}_c)^{\mathrm{T}} \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_c) \xrightarrow{\Sigma = \mathbf{I}} \hat{y} = \underset{c}{\operatorname{argmin}} (\boldsymbol{x} - \boldsymbol{\mu}_c)^{\mathrm{T}} (\boldsymbol{x} - \boldsymbol{\mu}_c)$$

$$\hat{y} = \underset{c}{\operatorname{argmin}}\, d^2(\boldsymbol{x}, \boldsymbol{\mu}_c)$$

- 基于高斯判别分析得出的距离度量方法，有较强的先验假定，比如给定类别的数据服从高斯分布
  - 是否可以不考虑数据分布假定，直接从数据中学习出度量距离的方法
- 距离度量学习
  - 从训练数据中学习出一个最适合相应任务的距离度量$d(\boldsymbol{x}, \boldsymbol{y})$，以期在该任务上达到最好的效果
  - 在数据表示足够好（基于深度学习的表示学习）情况下，基于学出来的距离度量的最近类中心点分类器可媲美很多复杂的分类器
  - 最近类平均度量学习（nearest class mean metric learning）

- 距离度量模型

$$d^2(\boldsymbol{x},\boldsymbol{\mu}_c)=\|\boldsymbol{x}_n-\boldsymbol{\mu}_c\|_{\mathbf{W}}^2=\left(\mathbf{W}(\boldsymbol{x}_n-\boldsymbol{\mu}_c)\right)^{\mathrm{T}}\left(\mathbf{W}(\boldsymbol{x}_n-\boldsymbol{\mu}_c)\right)=\|\mathbf{W}(\boldsymbol{x}_n-\boldsymbol{\mu}_c)\|_2^2$$

- 后验分布
  - –注意：这是判别模型

$$p(y=c|\boldsymbol{x};\boldsymbol{\mu},\mathbf{W})=\frac{\exp\left(-\frac{1}{2}\|\mathbf{W}(\boldsymbol{x}_n-\boldsymbol{\mu}_c)\|_2^2\right)}{\sum_{i=1}^{C}\exp\left(-\frac{1}{2}\|\mathbf{W}(\boldsymbol{x}_n-\boldsymbol{\mu}_i)\|_2^2\right)}$$

  - –可通过梯度下降算法求解最优值，具体可参见后续“logistic回归”课程

Fisher线性判别分析

- 高斯判别分析模型不适合处理高维数据
  - 数据维度D太大，容易造成估计的协方差矩阵非良态（不可逆）
  - 通过参数共享，对角化等间接手段来解决

- 解决方案
  - 间接方案
    - ✓协方差矩阵参数共享，对角化
  - 直接方案
    - ✓通过降维技术，根据一定的约束条件，把要处理的数据从D维空间映射到K维空间（$K \ll D$）
    - ✓典型方法： Fisher线性判别分析（Fisher linear discriminant analysis)

Fisher线性判别分析

- 考虑两分类问题，
  - 训练数据集$\mathcal{D} = \{(\boldsymbol{x}_n, y_n)\}_{n=1}^N$，$\boldsymbol{x}_n \in \mathbb{R}^D$, $y_n \in \{1, 2\}$
- 类条件均值向量

$$\boldsymbol{\mu}_1 = \frac{1}{N_1} \sum_{n:y_n=1} \boldsymbol{x}_n \qquad \boldsymbol{\mu}_2 = \frac{1}{N_2} \sum_{n:y_n=2} \boldsymbol{x}_n$$

- 令$\boldsymbol{w}$为拟求解的最优投影方向，则类条件均值向量在$\boldsymbol{w}$的投影为$m_k = \boldsymbol{w}^{\mathrm{T}} \boldsymbol{\mu}_k$，$\boldsymbol{x}_n$在$\boldsymbol{w}$的投影为$z_n = \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n$
- 类投影方差

$$s_k^2 = \sum_{n:y_n=k} (z_n - m_k)^2$$

- 目标函数

$$J(\boldsymbol{w}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2}$$

- 目标：求解$\boldsymbol{w}^*$，使得

$$\boldsymbol{w}^* = \underset{\boldsymbol{w}}{\operatorname{argmax}} J(\boldsymbol{w}) = \underset{\boldsymbol{w}}{\operatorname{argmax}} \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2}$$

  – 希望投影到$\boldsymbol{w}^*$上，两类的均值（中心点）之间的距离尽可能地远。

  – 希望投影到$\boldsymbol{w}^*$上，每类的类内方差尽可能地小。即，同类数据投影到$\boldsymbol{w}^*$上尽可能聚集在一起

- $J(\boldsymbol{w})$可改写成

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^{\mathrm{T}} \mathbf{S}_B \boldsymbol{w}}{\boldsymbol{w}^{\mathrm{T}} \mathbf{S}_V \boldsymbol{w}}$$

$$\mathbf{S}_B = (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^{\mathrm{T}}$$

$$\mathbf{S}_V = \sum_{n:y_n=1} (\boldsymbol{x}_n - \boldsymbol{\mu}_1)(\boldsymbol{x}_n - \boldsymbol{\mu}_1)^{\mathrm{T}} + \sum_{n:y_n=2} (\boldsymbol{x}_n - \boldsymbol{\mu}_2)(\boldsymbol{x}_n - \boldsymbol{\mu}_2)^{\mathrm{T}}$$

Fisher线性判别分析

$$\max J(\boldsymbol{w}) = \frac{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}}{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w}}$$

$$\frac{\partial}{\partial \boldsymbol{w}}\left(\frac{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}}{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w}}\right) = 0$$

$$\frac{1}{(\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w})^2}\left(\frac{\partial \boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}}{\partial \boldsymbol{w}}\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w} - \boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}\frac{\partial \boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w}}{\partial \boldsymbol{w}}\right) = 0$$

$$\frac{\partial \boldsymbol{x}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}}{\partial \boldsymbol{x}} = \left(\mathbf{A} + \mathbf{A}^{\mathrm{T}}\right)\boldsymbol{x}$$

$$\left(\left(\mathbf{S}_B + \mathbf{S}_B^{\mathrm{T}}\right)\boldsymbol{w}\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w} - \boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}\left(\mathbf{S}_V + \mathbf{S}_V^{\mathrm{T}}\right)\boldsymbol{w}\right) = 0$$

$$2\mathbf{S}_B\boldsymbol{w}\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w} = 2\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}\mathbf{S}_V\,\boldsymbol{w}$$

$$\mathbf{S}_B\boldsymbol{w} = \frac{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}}{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w}}\mathbf{S}_V\boldsymbol{w}$$

Fisher线性判别分析

$$\mathbf{S}_B\boldsymbol{w} = \frac{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}}{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w}}\mathbf{S}_V\boldsymbol{w}$$

$$\lambda = \frac{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_B\boldsymbol{w}}{\boldsymbol{w}^{\mathrm{T}}\mathbf{S}_V\boldsymbol{w}}$$

$$\mathbf{S}_B\boldsymbol{w} = \lambda\mathbf{S}_V\boldsymbol{w}$$

假定$\mathbf{S}_V$可逆

$$\mathbf{S}_V^{-1}\mathbf{S}_B\boldsymbol{w} = \lambda\boldsymbol{w}$$

$$\begin{aligned}\mathbf{S}_B\boldsymbol{w} &= (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^{\mathrm{T}}\boldsymbol{w} \\ &= (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(m_2 - m_1)\end{aligned}$$

$$\lambda\boldsymbol{w} = \mathbf{S}_V^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(m_2 - m_1)$$

$$\boldsymbol{w} \propto \mathbf{S}_V^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$$

只关注投影方向

$$\boldsymbol{w} = \mathbf{S}_V^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)$$

对$C(>2)$类分类可类似处理

不足：只能降到$K\ (\leq C-1)$维

Logistic回归